

Build Instructions — Windows Standalone

This guide walks through compiling **Space Shooter** into a standalone Windows executable (`.exe`).

1. Prerequisites

- **Unity Hub** installed.
 - **Unity 2022.3 LTS** editor installed (developed on `2022.3.40f1`).
 - During installation (or via Unity Hub → Installs → ⚙ → Add Modules) make sure “**Windows Build Support (Mono)**” and/or “**Windows Build Support (IL2CPP)**” are checked.
 - Mono builds faster and is fine for testing.
 - IL2CPP produces a faster, harder-to-decompile build (recommended for release; requires Visual Studio with the “Desktop development with C++” workload).
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2. Open the project

1. Launch **Unity Hub** → **Open** → **Add project from disk**.
 2. Select the `SpaceShooter/` folder (the one containing `Assets/` , `Packages/` , `ProjectSettings/`).
 3. Wait for the initial import and script compilation to finish. Confirm there are **no compiler errors** in the **Console** (red messages). Warnings are fine.
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3. Verify Build Settings scenes

1. Open **File** → **Build Settings...**
2. Under **Scenes In Build** you should see, in this order:
 - `Assets/Scenes/MainMenu.unity` → index **0**
 - `Assets/Scenes/GameOver.unity` → index **1**
3. If they are missing:
 - Open `MainMenu.unity` , then click **Add Open Scenes**.
 - Open `GameOver.unity` , then click **Add Open Scenes**.
 - Drag **MainMenu** above **GameOver** so MainMenu is index 0.
 - Tick the checkbox next to each so both are **enabled**.

The project ships with `ProjectSettings/EditorBuildSettings.asset` already listing both scenes, so this is usually pre-configured.

4. Select the target platform

1. Still in **Build Settings**, select **Windows, Mac, Linux** in the platform list.

2. If it is not already the active platform, click **Switch Platform** (this may take a moment).
3. Set:
 - **Target Platform:** `Windows`
 - **Architecture:** `Intel 64-bit (x86_64)`

5. (Optional) Player Settings

Open **Edit** → **Project Settings** → **Player** to customise:

- **Company Name / Product Name** (the product name becomes the `.exe` name).
- **Resolution and Presentation:** default windowed 1280×720 works well; enable Run In Background if desired.
- **Icon:** assign a custom icon if you have one.
- **Scripting Backend** (under Configuration): `Mono` (default) or `IL2CPP` (release).

6. Build

1. In **Build Settings**, click **Build** (or **Build And Run** to launch immediately).
2. Choose/create an **empty output folder**, e.g. `SpaceShooter/Build/Windows/`.
3. Unity compiles and writes the player. When finished you'll have:

```
Build/Windows/
├── SpaceShooter.exe          # the game executable
├── UnityPlayer.dll
├── SpaceShooter_Data/       # all game data
└── MonoBleedingEdge/ (Mono builds)
```

7. Run

- Double-click `SpaceShooter.exe`.
- To distribute, zip the **entire** output folder — the `.exe` requires the sibling `*_Data` folder and DLLs.

8. Command-line build (CI / automation, optional)

You can build headlessly with the Unity CLI. Example (adjust the Unity path and project path):

```
"C:\Program Files\Unity\Hub\Editor\2022.3.40f1\Editor\Unity.exe" ^
  -quit -batchmode -nologo ^
  -projectPath "C:\path\to\SpaceShooter" ^
  -buildWindows64Player "C:\path\to\SpaceShooter\Build\Windows\SpaceShooter.exe" ^
  -logFile "C:\path\to\SpaceShooter\build.log"
```

For full control (scenes, options) create an editor build script under `Assets/Editor/` that calls `BuildPipeline.BuildPlayer(...)` and invoke it with `-executeMethod`. This is optional — the GUI build

in
steps 4–6 is sufficient for most users.

9. Common issues

Symptom	Fix
“No scenes in build” / black screen	Re-add both scenes in Build Settings (step 3) and ensure they’re enabled.
Compiler errors on import	Confirm you’re on Unity 2022.3 LTS ; let the package manager finish resolving built-in modules.
Build button greyed out	Switch the platform to Windows first (step 4).
IL2CPP build fails	Install Visual Studio with the Desktop development with C++ workload, then rebuild.
<code>.exe</code> won’t start on another PC	Ship the whole output folder (the <code>_Data</code> folder + DLLs must accompany the exe).